Lab 08

ENVX2001 Applied Statistical Methods

Contents

Welcome to week 8! 2
Learning outcomes 2
Specific goals 3
Preparation 3
Packages 3
1. Variable selection for California streamflow (~ 45 mins) 4
. 5
Exercise 1 - exploring full and reduced models 5
Solution 6
(i) Read in the data and investigate the relationship between each of your variables. Describe the relationships, supported by values (correlation coefficient) and plots. 6
(ii) Fit a multiple linear regression model to the data using all variables. Set `runoff_volume` as the response variable and `lake_sabrina`, `pine_creek` and `rock_creek` as the predictor variables. This will be known as the 'full model'. . . . 8
(iii) Check the assumptions of your model. Do you need to transform any variables? What does the Variance Inflation Factor (VIF) tell you about the collinearity between predictor variables? 8
(iv) Make a note of the model parameters and fit statistics for the full model. . 13
(v) Remove `lake_sabrina` from the model and run the model again. This will be known as the 'reduced model'. 14
(vi) Is the reduced model an improvement on the full model? How can you tell? What statistics would you look at to determine this? 14
. 15
Exercise 2 - which model is a significant improvement? 15
Solution 15
(i) State the null and alternative hypotheses for the F-test between the full and reduced models. 15
(ii) Perform the F-test using the `anova()` function to compare the full and reduced models. What is the p-value? Based on this, what might we conclude to be the better model? 16
. 16
Exercise 3 - stepwise regression 16
Solution 16

(i) When using the AIC, how do we select the better model? 16
(ii) Have a go at running the `step()` function on the full model. Which model does it select as the best model? 17
(iii) One final check: what are the AIC values for the full and reduced models? Which model has the lowest AIC value? 17
(iv) Based on our tests, which model do you think is the best model to use for predicting runoff volume at Bishop, California? Why have you chosen this model? 18
2. Variable selection for plant aridity (~ 45 mins) 19
.... 20
Exercise 1 20
Solution 21
Conclusion 36
Closing thoughts 36
Attribution 36

Welcome to week 8!

This week's lab will give you some practice improving models by selecting the most useful predictors.

This will prepare you for next week's content where we will use models for prediction.

Learning outcomes

This module links to the following unit learning outcomes:

- LO3. demonstrate proficiency in the use or the statistical programming language R to apply an ANOVA and fit regression models to experimental data
- LO5. interpret the output and understand conceptually how its derived of a regression, ANOVA and multivariate analysis that have been calculated by R
- LO6. write statistical and modelling results as part of a scientific report

In this lab, we will learn how to:

1. Fit multiple linear regression models and interpret associated indicators of collinearity

2. Draw upon various methods to determine whether a full or reduced model is more suitable

Specific goals

By the end of this lab, you should be able to:

- ☐ Fit a multiple linear regression model
- ☐ Interpret VIF plots and output to determine whether collinearity is a concern in your model
- ☐ Remove non-significant predictors to produce a more parsimonious model
- ☐ Compare a full and reduced model using an F-test
- ☐ Run and interpret stepwise regression and associated output
- ☐ Compare a full and reduced model using the AIC
- ☐ Use the results of model comparisons to select the best model for your data

Preparation

Please work on these exercises by creating your own Quarto file. You are welcome to write your responses up in the style of a report.

Packages

This lab uses `tidyverse`, `readxl`, `moments`, `psych`, `performance`, `car`. Install any you are missing by running the following **in the console**:

```
CODE
install.packages(c("car", "tidyverse", "readxl", "moments", "psych", "performance"))
```

If a package is already installed, R will simply reinstall it — no harm done.

1. Variable selection for California streamflow (~ 45 mins)

Figure 1: Little Lakes Valley in the Eastern Sierra Nevada Mountains outside of Bishop, California. Source: Adobe Stock.

Last week you were investigating the relationship between precipitation at three sites (Lake Sabrina, Pine Creek, and Rock Creek) and stream runoff volume at a site near Bishop, California. You found that Lake Sabrina was not a significant predictor of runoff volume, but Pine Creek and Rock Creek were.

This week we will follow steps to see if we can produce a more parsimonious model. We will be focussing on the process rather than report writing, but we still encourage you to write up your conclusions as if it were for a report.

As a recap, the california_streamflow dataset you were analysing contains 43 years of annual precipitation measurements (in mm) taken at (originally) 6 sites in the Owens Valley in California. You will focus on only 3 variables.

The data is in the **california_streamflow.xlsx** file.

Variables are as follows:

- `lake_sabrina`: Precipitation recorded at Lake Sabrina (mm)
- `pine_creek`: Precipitation recorded at Big Pine Creek (mm),
- `rock_creek`: Precipitation recorded at Rock Creek (mm)
- `runoff_volume`: Stream runoff volume measured at a site near Bishop, California (ML/year). This will be the response variable.

Note the variables have already been log-transformed to increase normality of the residuals in the regressions.

Data source: Hollett, K.J., Danskin, F.M., McCaffrey, W.F., and Walti, S.L., 1991. Geology and water resources of Owens Valley, California. U.S. Geological Survey Water-Supply Paper 2370-H.

Exercise 1 - exploring full and reduced models

(i) Read in the data and investigate the relationship between each of your variables. Describe the relationships, supported by values (correlation coefficient) and plots.

(ii) Fit a multiple linear regression model to the data using all variables. Set `runoff_volume` as the response variable and `lake_sabrina`, `pine_creek` and `rock_creek` as the predictor variables. This will be known as the 'full model'.

(iii) Check the assumptions of your model. Do you need to transform any variables? What does the Variance Inflation Factor (VIF) tell you about the collinearity between predictor variables?

HINT: The VIF will appear as one of the plots in the `check_model()` function, but it can also be helpful to get the values. You can use the `vif()` function from the `car` package to get the VIF values for each predictor variable.

The check_model function has been squishing plots, so it may be better at times to plot them individually. You may use the following code, adapted to your model object names:

CODE

```r
library(performance)

# Linearity
check_model(full_model, check = c("linearity"))

# Normality
check_model(full_model, check = c("qq", "normality"))

# Equal variance
check_model(full_model, check = c("homogeneity"))

# Outliers
check_model(full_model, check = c("outliers", "pp_check"))
```

```
# VIF - visual
plot(performance::check_collinearity(full_model))

# VIF - numbers
car::vif(full_model)
```

(iv) Make a note of the model parameters and fit statistics for the full model.

(v) Remove `lake_sabrina` from the model and run the model again. This will be known as the 'reduced model'.

(vi) Is the reduced model an improvement on the full model? How can you tell? What statistics would you look at to determine this?

Solution

***(i)* Read in the data and investigate the relationship between each of your variables. Describe the relationships, supported by values (correlation coefficient) and plots.**

CODE

```
library(readxl)

# read in the data
stream_data <- read_xlsx("data/california_streamflow.xlsx", "streamflow")

library(tidyverse)
glimpse(stream_data) # check the data
```

OUTPUT

```
Rows: 43
Columns: 4
$ lake_sabrina  <dbl> 1.958717, 2.087881, 1.981175, 2.054169, 2.102934, 2.1568…
$ pine_creek    <dbl> 2.017618, 2.282781, 2.383471, 2.451719, 2.618086, 2.3532…
$ rock_creek    <dbl> 2.275823, 2.450548, 2.491194, 2.585246, 2.706948, 2.3159…
$ runoff_volume <dbl> 4.825412, 4.920867, 4.911735, 4.924242, 5.120841, 4.9210…
```

CODE

```
# Correlations and scatterplots all in one (+ histograms for each variable)
library(psych)
```

OUTPUT

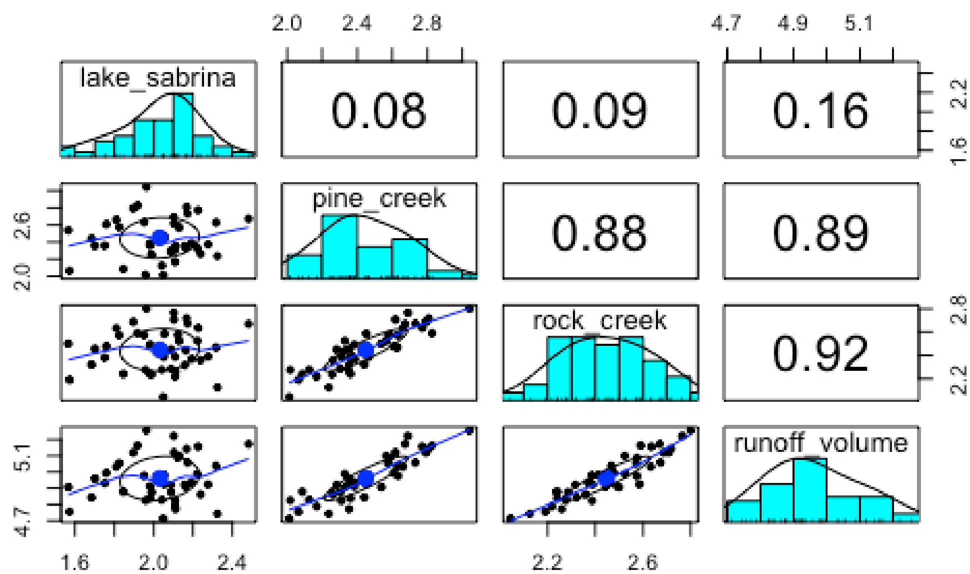
```

Attaching package: 'psych'
```

OUTPUT

```
The following objects are masked from 'package:ggplot2':

    %+%, alpha
```

CODE

```
pairs.panels(stream_data)
```



CODE

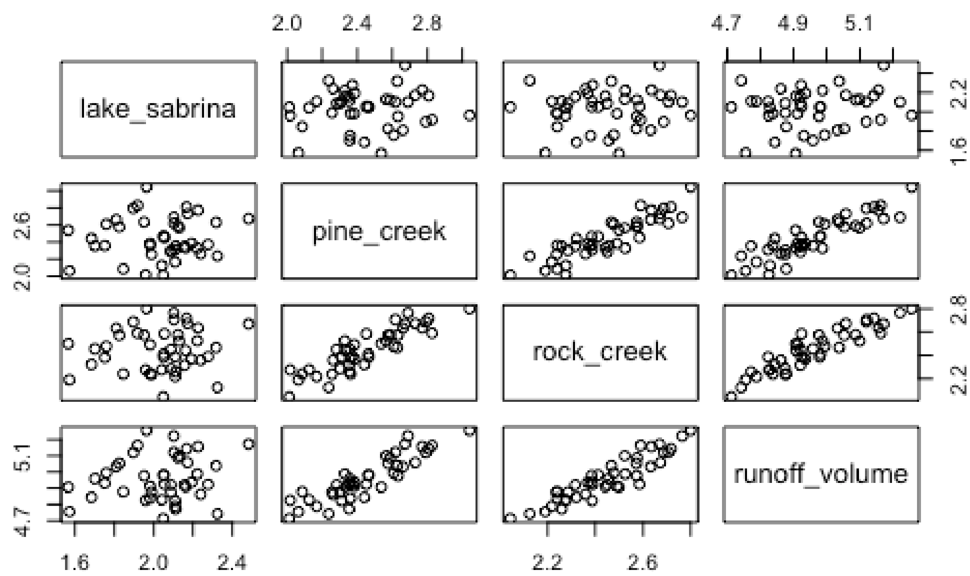
```
# can also get the same results separately
cor(stream_data)
```

OUTPUT

```
              lake_sabrina pine_creek rock_creek runoff_volume
lake_sabrina    1.00000000 0.07848517 0.09054996     0.1649284
pine_creek      0.07848517 1.00000000 0.88381452     0.8929251
rock_creek      0.09054996 0.88381452 1.00000000     0.9194688
runoff_volume   0.16492838 0.89292511 0.91946880     1.0000000
```

CODE

```
pairs(stream_data)
```



Similar to what we found last week, we see that runoff volume has a strong relationship with rock creek and pine creek, but not with lake sabrina. We also see that there is a strong relationship between rock creek and pine creek, which could be a sign of collinearity. We can confirm if this may be an issue with the model by observing the Variance inflation factor (VIF) later on when we check our model assumptions.

***(ii)* Fit a multiple linear regression model to the data using all variables. Set `runoff_volume` as the response variable and `lake_sabrina`, `pine_creek` and `rock_creek` as the predictor variables. This will be known as the 'full model'.**

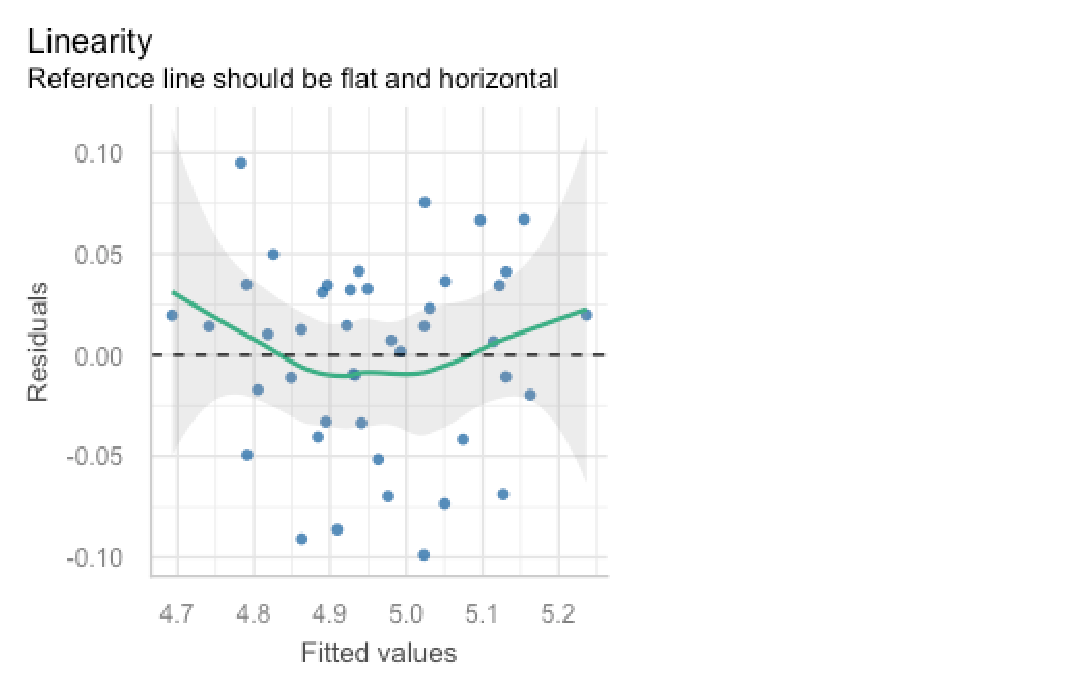
CODE
```
# fit the full model

full_model <- lm(runoff_volume ~ lake_sabrina + pine_creek + rock_creek, data = stream_data)
```

***(iii)* Check the assumptions of your model. Do you need to transform any variables? What does the Variance Inflation Factor (VIF) tell you about the collinearity between predictor variables?**

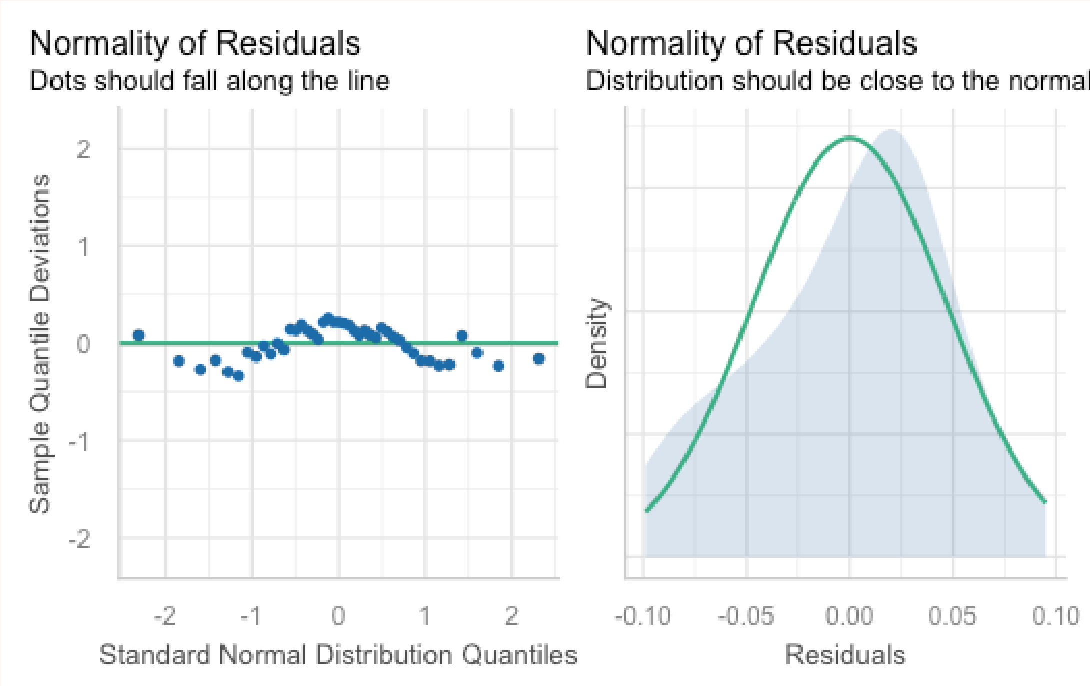
HINT: The VIF will appear as one of the plots in the `check_model()` function, but it can also be helpful to get the values. You can use the `vif()` function from the `car` package to get the VIF values for each predictor variable.

The check_model function has been squishing plots, so it may be better at times to plot them individually. You may use the following code, adapted to your model object names:

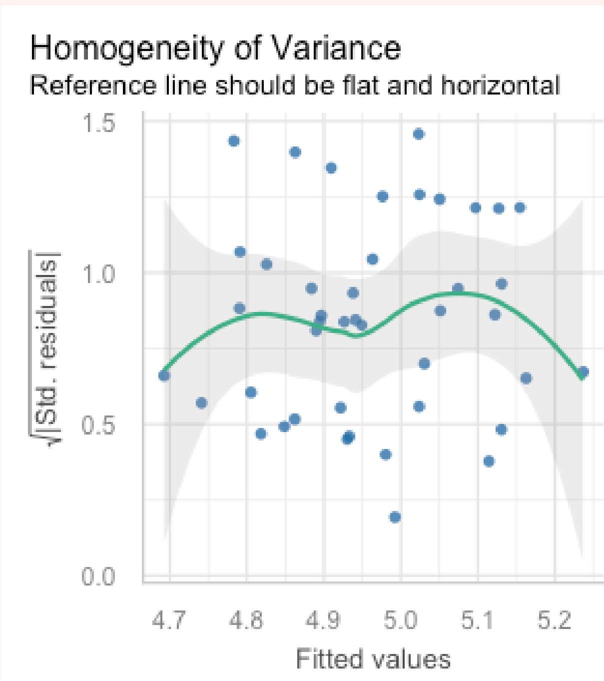
```
CODE
library(performance)

# Linearity
check_model(full_model, check = c("linearity"))
```




```
CODE
# Normality
check_model(full_model, check = c("qq", "normality"))
```

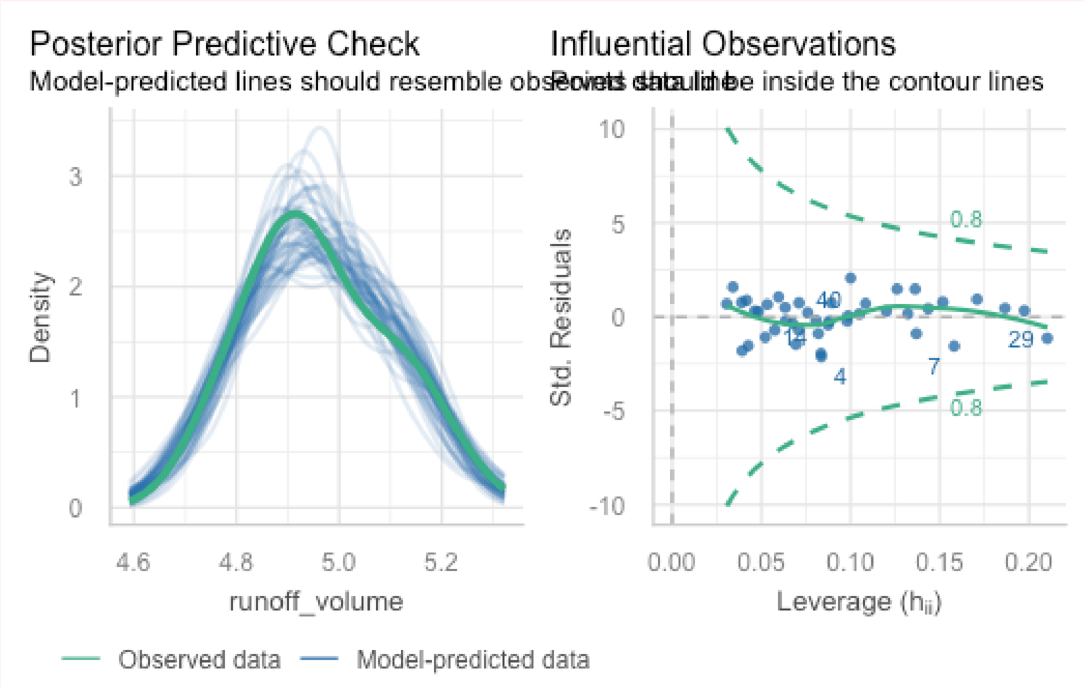



```
CODE
# Equal variance
check_model(full_model, check = c("homogeneity"))
```



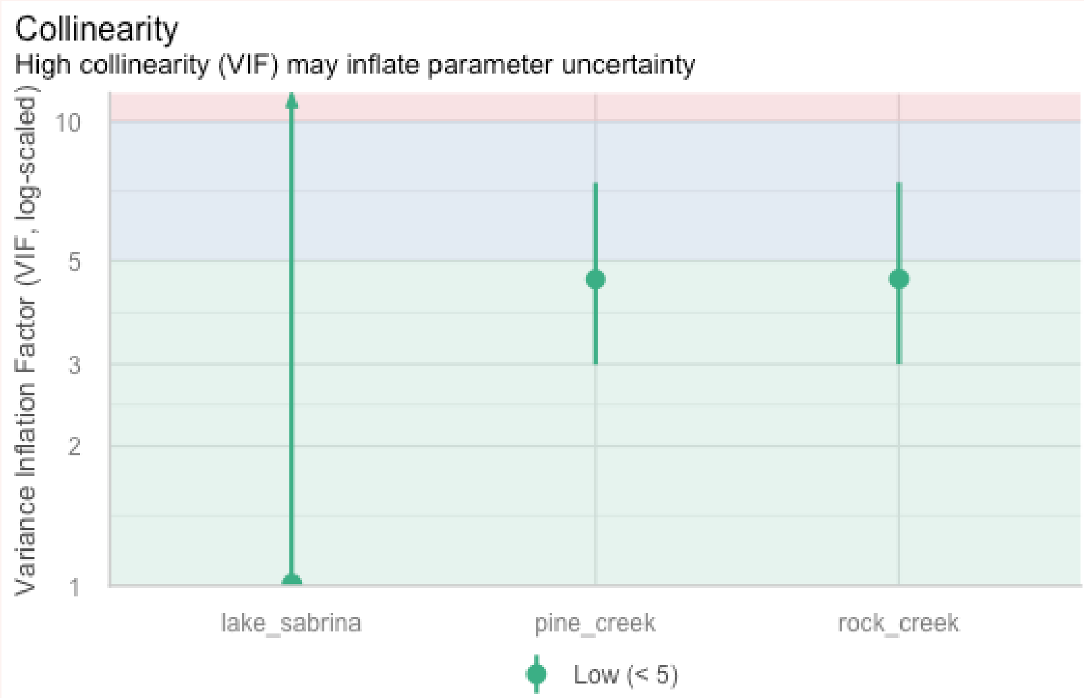
CODE

```
# Outliers
check_model(full_model, check = c("outliers", "pp_check"))
```



CODE

```
# VIF - visual
plot(performance::check_collinearity(full_model))
```



CODE

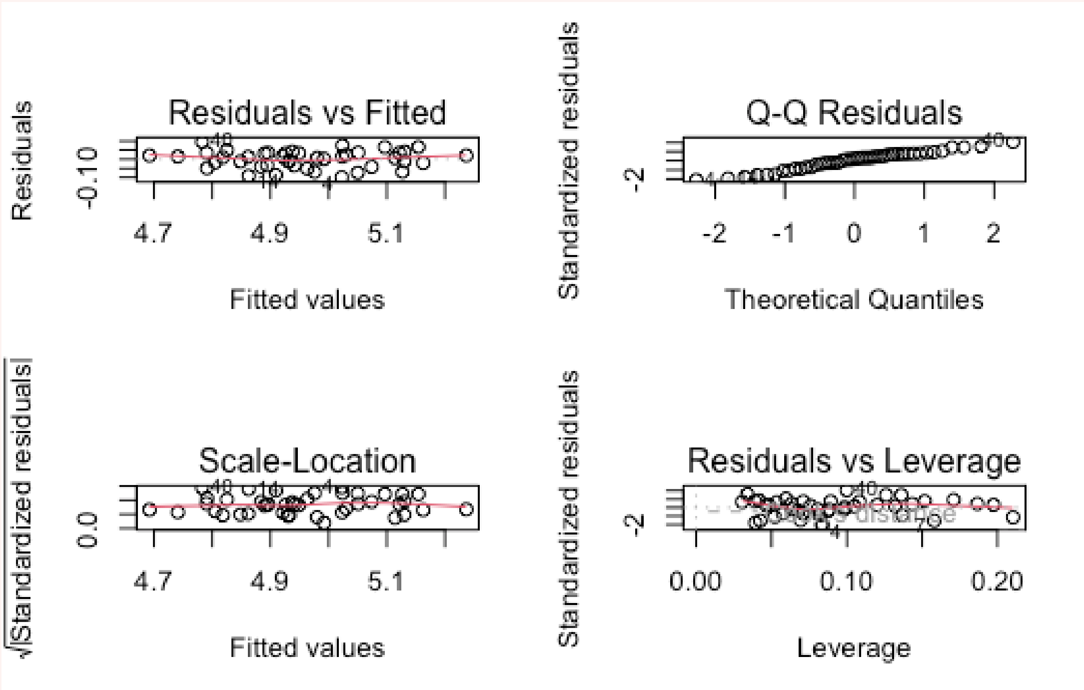
```
# VIF - numbers
car::vif(full_model)
```

OUTPUT

```
lake_sabrina   pine_creek   rock_creek
    1.008278     4.568933     4.578328
```

CODE

```
#### ALTERNATIVE: plot() function to check assumptions

par(mfrow=c(2,2))

plot(full_model)
```



CODE
```
par(mfrow=c(1,1))
```

Lake Sabrina = 1.01

Pine_creek = 4.57

Rock_creek = 4.58

Our threshold is 5, so given all of our VIFs are below this, we are not too concerned about collinearity. However, because they are close to the threshold it is something to keep in mind when interpreting the model parameters.

(iv) **Make a note of the model parameters and fit statistics for the full model.**

CODE
```
summary(full_model)
```

OUTPUT
```

Call:
lm(formula = runoff_volume ~ lake_sabrina + pine_creek + rock_creek,
    data = stream_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.09885 -0.03331  0.01025  0.03359  0.09495 

Coefficients:
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.25716    0.12360  26.352  < 2e-16 ***
lake_sabrina  0.05631    0.03756   1.499  0.14185
pine_creek    0.21085    0.06756   3.121  0.00339 **
rock_creek    0.43838    0.08798   4.983 1.32e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04861 on 39 degrees of freedom
Multiple R-squared:  0.8817,	Adjusted R-squared:  0.8726
F-statistic: 96.88 on 3 and 39 DF,  p-value: < 2.2e-16
```

As we saw last week, Lake Sabrina is not a significant predictor of runoff volume, but Pine Creek and Rock Creek are. The model has an adjusted R^2 of 0.87, which means that 87% of the variation in runoff volume can be explained by the three predictor variables.

***(v)* Remove `lake_sabrina` from the model and run the model again. This will be known as the 'reduced model'.**

CODE

```
reduced_model <- lm(runoff_volume ~ pine_creek + rock_creek, data = stream_data)
```

***(vi)* Is the reduced model an improvement on the full model? How can you tell? What statistics would you look at to determine this?**

CODE

```
summary(full_model)
```

OUTPUT

```
Call:
lm(formula = runoff_volume ~ lake_sabrina + pine_creek + rock_creek,
    data = stream_data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.09885 -0.03331  0.01025  0.03359  0.09495

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.25716    0.12360  26.352  < 2e-16 ***
lake_sabrina  0.05631    0.03756   1.499  0.14185
pine_creek    0.21085    0.06756   3.121  0.00339 **
rock_creek    0.43838    0.08798   4.983 1.32e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04861 on 39 degrees of freedom
Multiple R-squared:  0.8817,	Adjusted R-squared:  0.8726
F-statistic: 96.88 on 3 and 39 DF,  p-value: < 2.2e-16
```

CODE

```
summary(reduced_model)
```

OUTPUT

```
Call:
lm(formula = runoff_volume ~ pine_creek + rock_creek, data = stream_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.09832 -0.02350  0.01076  0.03291  0.08568 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  3.35762    0.10547  31.835  < 2e-16 ***
pine_creek   0.21051    0.06861   3.068  0.00385 ** 
rock_creek   0.44437    0.08925   4.979 1.26e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04937 on 40 degrees of freedom
Multiple R-squared:  0.8749,	Adjusted R-squared:  0.8686 
F-statistic: 139.8 on 2 and 40 DF,  p-value: < 2.2e-16
```

Removing the `lake_sabrina` predictor from the model has slightly decreased the adjusted R^2 value, but it has also removed a non-significant predictor from the model. This is an improvement in terms of parsimony, but we will need to use hypothesis testing to determine whether the reduced model is a *significant* improvement on the full model. We can do this using an F-test, which we will explore in the next exercise.

Exercise 2 - which model is a significant improvement?

We can compare the models using the adjusted r^2, but we will need to use hypothesis testing to decide whether the reduced model is a significant improvement or not. Let's explore this by using the F-test to compare the full and reduced models.

(i) State the null and alternative hypotheses for the F-test between the full and reduced models.

(ii) Perform the F-test using the `anova()` function to compare the full and reduced models. What is the p-value? Based on this, what might we conclude to be the better model?

Solution

***(i)* State the null and alternative hypotheses for the F-test between the full and reduced models.**

H0: No significant difference between full and reduced models

H1: There is a significant difference between full and reduced models *(ii)* **Perform the F-test using the `anova()` function to compare the full and reduced models. What is the p-value? Based on this, what might we conclude to be the better model?**

CODE

```r
anova(full_model, reduced_model)
```

OUTPUT

```
Analysis of Variance Table

Model 1: runoff_volume ~ lake_sabrina + pine_creek + rock_creek
Model 2: runoff_volume ~ pine_creek + rock_creek
  Res.Df      RSS Df  Sum of Sq      F Pr(>F)
1     39 0.092166
2     40 0.097478 -1 -0.0053121 2.2478 0.1419
```

The P-value is 0.1419, which is larger than 0.05, so we fail to reject the null hypothesis.

This means that the reduced model is not significantly different to the full model, so we would opt for the reduced model as it is more parsimonious.

Exercise 3 - stepwise regression

Instead of removing one predictor at a time, there is a much quicker way involving the `step()` function in R. This function performs stepwise regression, which is an automated process of adding or removing predictors based on their significance in the model and the Akaike Information Criterion (AIC).

(i) When using the AIC, how do we select the better model?

(ii) Have a go at running the `step()` function on the full model. Which model does it select as the best model?

(iii) One final check: what are the AIC values for the full and reduced models? Which model has the lowest AIC value?

HINT: you can use the `AIC()` function to get the AIC values for each model. Insert the model name between the brackets.

(iv) Based on our tests, which model do you think is the best model to use for predicting runoff volume at Bishop, California? Why have you chosen this model?

Solution

(i) **When using the AIC, how do we select the better model?**
Select the model with the lowest AIC value.

The AIC is a measure of the relative quality of a statistical model for a given set of data. It takes into account the goodness of fit and the complexity of the model (number of predictors). A lower AIC value indicates a better model.

(ii) **Have a go at running the `step()` function on the full model. Which model does it select as the best model?**

CODE

```
step_model <- step(full_model, direction = "backward")
```

OUTPUT

```
Start:  AIC=-256.25
runoff_volume ~ lake_sabrina + pine_creek + rock_creek

               Df Sum of Sq      RSS     AIC
<none>                      0.092166 -256.25
- lake_sabrina  1  0.005312 0.097478 -255.84
- pine_creek    1  0.023015 0.115181 -248.66
- rock_creek    1  0.058680 0.150845 -237.07
```

The `step()` function selects the full model as the best model, which is inconsistent with our previous analysis using the F-test.

This is because the `step()` function uses the AIC to compare models, which is a different criterion than the F-test. The AIC takes into account both the goodness of fit and the complexity of the model, while the F-test only compares the goodness of fit between two models.

In this case, the AIC may be favoring the full model due to its better fit, even though it includes a non-significant predictor.

(iii) **One final check: what are the AIC values for the full and reduced models? Which model has the lowest AIC value?**
HINT: you can use the `AIC()` function to get the AIC values for each model. Insert the model name between the brackets.

CODE

```
AIC(full_model)
```

OUTPUT

```
[1] -132.2222
```

CODE

```
AIC(reduced_model)
```

OUTPUT

```
[1] -131.8126
```

We can see that the full model has the lowest AIC value.

(iv) **Based on our tests, which model do you think is the best model to use for predicting runoff volume at Bishop, California? Why have you chosen this model?**
The reduced model is better. It is more parsimonious and does not include a non-significant predictor.

The F-test also indicates that the reduced model is not a significant improvement on the full model, so we would opt for the reduced model as it is simpler and just as good at explaining the variation in runoff volume.

We did see that the step function selected the full model as the best model based on AIC, as did our separate AIC calculation, but we would be cautious about this result given that it includes a non-significant predictor and the F-test did not support it as a significant improvement.

Automated model selection methods are helpful when you have a large number of predictors, and calculating the AIC can assist with model selection, but our results demonstrate why we should not rely solely on these methods. We should also consider the significance of predictors and the overall fit of the model when making decisions about which model to use.

Tip

Before moving on to the next section, take a short break and stretch your legs.

2. Variable selection for plant aridity (~ 45 mins)

Figure 2: Image of invasive capeweed. Source: Adobe Stock.

This dataset examines the adaptation of invasive capeweed (*Arctotheca calendula*) populations to aridity gradients across southern Australia. A common garden experiment was conducted from 2018-2019 with populations sourced from varying aridity levels, subjecting them to wet and dry treatments to measure traits related to drought escape, avoidance, and tolerance.

The dataset has been reduced to contain only the variables you will need to fit in the model. Furthermore, only the samples that received the wet treatment will be used for this analysis (n = 89 observations).

The dataset is in the *plant_aridity.csv* file.

Variables are as follows:

- `growth_rate`: growth rate measured as differences in # of rosette leaves between zero and eight weeks after transplanting
- `days_to_flower`: days to first flowering since transplant (days)
- `specific_leaf_area`: specific leaf area as [leaf area (cm2) / dry mass (g)]
- `root_shoot_ratio`: root-to-shoot ratio [root mass/shoot mass]
- `chlorogenic_acid`: concentration of chlorogenic acid (mg/g frwt sample)
- `rel_fitness`: Relative fitness of each plant, seed count divided by the mean seed count

`rel_fitness` will be the response variable in your models, and the other variables will be the predictor variables.

Data source: Uesugi, Akane (2022). Data from: Multivariate selection mediated by aridity predicts divergence of drought resistant traits along natural aridity gradients of an invasive weed [Dataset]. Dryad. https://doi.org/10.5061/dryad.tht76hf17

Paper: Carvalho C, Davis R, Connallon T, Gleadow RM, Moore JL, Uesugi A. Multivariate selection mediated by aridity predicts divergence of drought-resistant traits along natural aridity gradients of an invasive weed. New Phytol. 2022 May;234(3):1088-1100. doi: 10.1111/nph.18018.

> **Exercise 1**
>
> *(i)* Write out the statistical model for a multiple linear regression and define it in terms of the data (i.e. relative fitness vs all other variables).
>
> *(ii)* State the hypotheses you will be testing for the relationship between each variable and relative fitness.
>
> *(iii)* Read in the data and conduct some exploratory data analysis upon your variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.
>
> *(iv)* Investigate the relationship between each of your variables. Describe the relationship, supported by values (correlation coefficient) and plots.
>
> Hint: we are not only looking at relationship between response and predictors here, but also the relationships between predictor variables (signs of collinearity).
>
> *(v)* Fit a multiple linear regression model to the data, with `rel_fitness` as the response variable and `growth_rate`, `days_to_flower`, `specific_leaf_area`, `root_shoot_ratio`, and `chlorogenic_acid` as the predictor variables. Label this model as `full_plant_model`.
>
> *(vi)* Check the assumptions of your model. Do you need to transform any variables?
>
> Hint 1: you can use the the `check_model()` function from the `performance` package to check the assumptions of your model. Because the performance package can squish plots and

make them hard to interpret, you can specify plots of interest, as shown in the previous exercises.

Hint 2: If you are unsure about the collinearity plot, you can use the `vif()` function from the `car` package to get the VIF values for each predictor variable.

Hint 3: If you find that the assumptions are not met you can use log-transformations or other transformations as appropriate based on the distribution of your variables. If you are log-transforming a variable which contains negative or zero values you will need to add a constant, i.e. `log(x+1)`

(vii) Interpret the model parameters and fit statistics. Are there any significant predictors of relative fitness? does the model explain much variation in relative fitness?

(viii) Remove any highly non-significant predictors (P > 0.1) from the model and run the model again. This will be known as the 'reduced model'.

(ix) Run a partial f-test to determine whether the reduced model an improvement on the full model. Which model is better?

(x) Run stepwise regression using the `step()` function fitted to the full model. Which model does it select as the best model?

(viii) Write a concluding statement on your findings.

Solution

(i) Write out the statistical model for a multiple linear regression and define it in terms of the data (i.e. relative fitness vs all other variables).

You can write it out in words and/or in mathematical notation.

$$\begin{aligned}\text{relative\_fitness} = & \beta_0 + \beta_1 \cdot \text{growth\_rate} + \beta_2 \cdot \text{days\_to\_flower} \\ & + \beta_3 \cdot \text{specific\_leaf\_area} + \beta_4 \cdot \text{root\_shoot\_ratio} \\ & + \beta_5 \cdot \text{chlorogenic\_acid} + \epsilon_i\end{aligned}$$

(ii) State the hypotheses you will be testing for the relationship between each variable and relative fitness.

In multiple linear regression we are testing multiple hypotheses (one per predictor + one for the overall model).

Therefore:

1. Relationship between growth_rate and relative_fitness:

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

2. days_to_flower and relative_fitness:

$$H_0 : \beta_2 = 0$$

$$H_1 : \beta_2 \neq 0$$

3. specific_leaf_area and relative_fitness:

$$H_0 : \beta_3 = 0$$

$$H_1 : \beta_3 \neq 0$$

4. root_shoot_ratio and relative_fitness:

$$H_0 : \beta_4 = 0$$

$$H_1 : \beta_4 \neq 0$$

5. chlorogenic_acid and relative_fitness:

$$H_0 : \beta_5 = 0$$

$$H_1 : \beta_5 \neq 0$$

4. Overall model:

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_1 : \text{at least one} \ \ \beta_i \neq 0$$

(iii) Read in the data and conduct some exploratory data analysis upon your variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.

CODE
```
plant_aridity <- read.csv("data/plant_aridity.csv")

str(plant_aridity)
```

OUTPUT
```
'data.frame':   89 obs. of  6 variables:
 $ growth_rate      : num  0.221 1.436 0.309 0.221 1.613 ...
 $ days_to_flower   : num  -0.415 -0.627 -0.202 0.859 -0.839 ...
 $ root_shoot_ratio  : num  1.207 1.204 0.961 0.652 0.633 ...
 $ specific_leaf_area: num  0.248 0.489 0.793 0.461 -0.247 ...
 $ chlorogenic_acid  : num  0.0614 2.8136 -0.3105 -1.7247 -0.5706 ...
 $ rel_fitness       : num  1.254 0.682 1.355 2.436 1.61 ...
```

Because we have more variables, it is ok to use the summary function here and report the values.

CODE

```r
summary(plant_aridity)
```

OUTPUT

```
  growth_rate       days_to_flower     root_shoot_ratio  specific_leaf_area
 Min.   :-1.51639   Min.   :-1.90082   Min.   :-0.9722   Min.   :-2.39281
 1st Qu.: 0.04853   1st Qu.:-0.83922   1st Qu.:-0.7796   1st Qu.:-0.23729
 Median : 0.57017   Median :-0.20226   Median :-0.5712   Median : 0.06985
 Mean   : 0.53331   Mean   : 0.06731   Mean   :-0.4158   Mean   : 0.08423
 3rd Qu.: 1.09181   3rd Qu.: 0.64702   3rd Qu.:-0.2839   3rd Qu.: 0.49105
 Max.   : 1.78559   Max.   : 2.98253   Max.   : 1.2066   Max.   : 2.44174
 chlorogenic_acid      rel_fitness
 Min.   :-2.196e+00   Min.   :0.6817
 1st Qu.:-5.706e-01   1st Qu.:1.3553
 Median : 2.883e-02   Median :1.8132
 Mean   : 6.454e-05   Mean   :1.8667
 3rd Qu.: 7.222e-01   3rd Qu.:2.2589
 Max.   : 2.814e+00   Max.   :4.3504
```

CODE

```r
apply(plant_aridity, 2, sd) # apply sd over columns
```

OUTPUT

```
       growth_rate     days_to_flower   root_shoot_ratio specific_leaf_area
         0.7133256          1.0508145          0.5095113          0.6444291
  chlorogenic_acid        rel_fitness
         1.0123975          0.6384944
```

CODE

```r
moments::skewness(plant_aridity, na.rm = TRUE)
```

OUTPUT

```
       growth_rate     days_to_flower   root_shoot_ratio specific_leaf_area
        -0.3885121          0.3767381          1.3954288         -0.8651712
  chlorogenic_acid        rel_fitness
        -0.1603996          0.7169594
```

Growth rate: mean growth rate is 0.533, which is similar to the median (0.570), suggesting a relatively symmetric distribution. The standard deviation is 0.713. The skewness is −0.389, which is close to 0, indicating a very symmetric distribution.

Days to flower: mean is 0.067, which is close to the median (-0.202). The standard deviation is 1.051. The skewness is 0.377, which is close to zero, indicating a very symmetric distribution.

root_shoot_ratio: mean is −0.416, which is not too close to the median (-0.571), suggesting a possibly skewed distribution. The standard deviation is 0.510. The skewness is 1.395, which is greater than 1, indicating a positively skewed distribution. It may be worth transforming this variable to see if a relationship becomes apparent.

specific_leaf_area: mean is 0.084, which is close to the median (0.069). The standard deviation is 0.644. The skewness is −0.865, which is between −0.5 and −0.1, indicating a degree of skewness.

chlorogenic_acid: mean is 0.000006454, which is close to the median (0.00283). The standard deviation is 1.012. The skewness is −0.160, which is close to zero, indicating a very symmetric distribution.

rel_fitness: mean is 1.867, which is close to the median (1.813), suggesting a symmetric distribution. The standard deviation is 0.070. The skewness is 0.717, which is indicating a low amount of skewness.

Boxplots indicate the at there are outliers present within each variable, but we can address the leverage plot when we go to test our model assumptions.

CODE

```
library(gridExtra)
```

OUTPUT

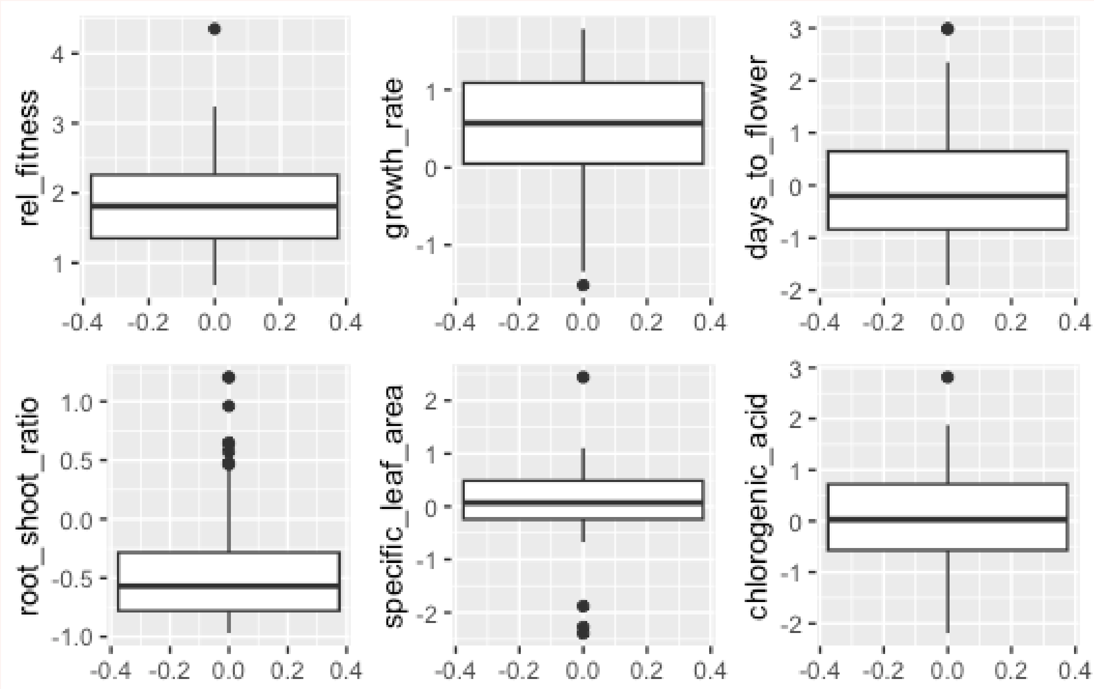
```

Attaching package: 'gridExtra'
```

OUTPUT

```
The following object is masked from 'package:dplyr':

    combine
```

CODE

```
grid.arrange(
    ggplot(plant_aridity, aes(y = rel_fitness)) + geom_boxplot(),
    ggplot(plant_aridity, aes(y = growth_rate)) + geom_boxplot(),
    ggplot(plant_aridity, aes(y = days_to_flower)) + geom_boxplot(),
    ggplot(plant_aridity, aes(y = root_shoot_ratio)) + geom_boxplot(),
    ggplot(plant_aridity, aes(y = specific_leaf_area)) + geom_boxplot(),
    ggplot(plant_aridity, aes(y = chlorogenic_acid)) + geom_boxplot(),
    nrow = 2
)
```



(iv) Investigate the relationship between each of your variables. Describe the relationship, supported by values (correlation coefficient) and plots.

Hint: we are not only looking at relationship between response and predictors here, but also the relationships between predictor variables (signs of collinearity).

CODE

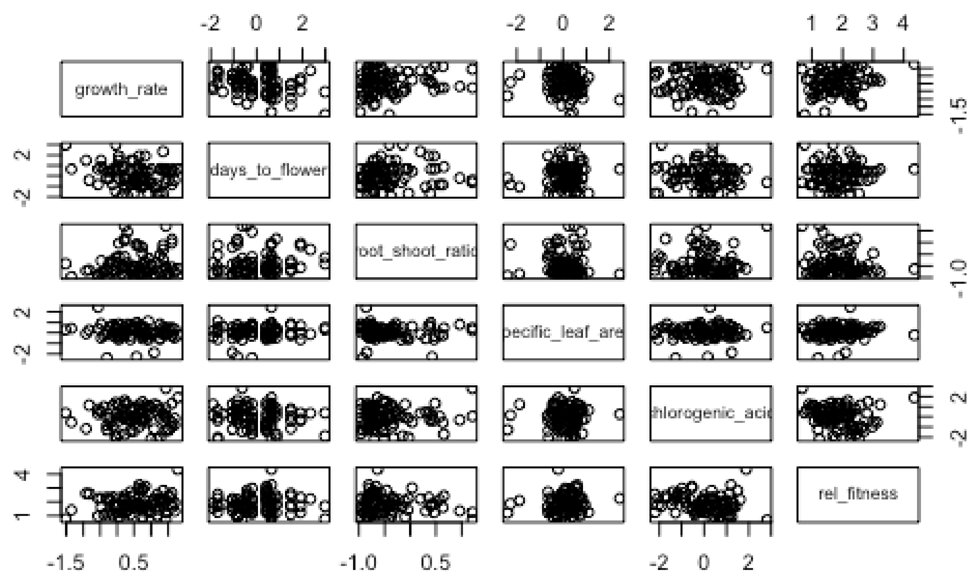
```
cor(plant_aridity)
```

OUTPUT

```
                   growth_rate days_to_flower root_shoot_ratio
growth_rate         1.00000000    -0.21623517       0.13604063
days_to_flower     -0.21623517     1.00000000       0.04914926
root_shoot_ratio    0.13604063     0.04914926       1.00000000
specific_leaf_area -0.12589551     0.08133447      -0.05095794
chlorogenic_acid    0.01249025    -0.08041919       0.04684624
rel_fitness         0.18208449     0.05277464      -0.13799422
                   specific_leaf_area chlorogenic_acid rel_fitness
growth_rate               -0.12589551       0.01249025  0.18208449
days_to_flower             0.08133447      -0.08041919  0.05277464
root_shoot_ratio          -0.05095794       0.04684624 -0.13799422
specific_leaf_area         1.00000000       0.02726978  0.10073474
chlorogenic_acid           0.02726978       1.00000000 -0.16645089
rel_fitness                0.10073474      -0.16645089  1.00000000
```

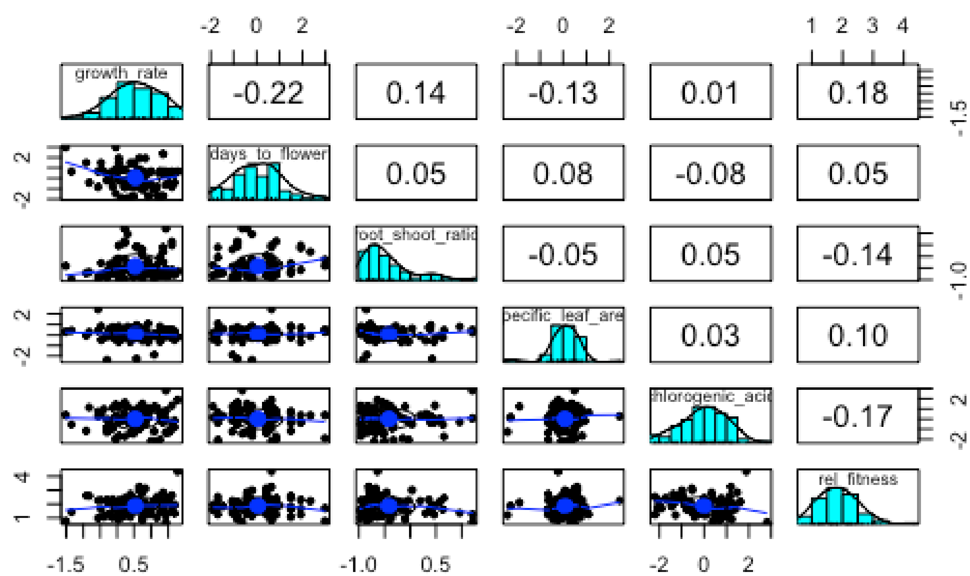
CODE

```
pairs(plant_aridity)
```



CODE

```
pairs.panels(plant_aridity)
```



Low correlation coefficients (r < 0.2) between all variables, but the correlations are stronger between each variable and rel_fitness.

(v) Fit a multiple linear regression model to the data, with `rel_fitness` as the response variable and `growth_rate`, `days_to_flower`, `specific_leaf_area`, `root_shoot_ratio`, and `chlorogenic_acid` as the predictor variables. Label this model as `full_plant_model`.

CODE

```
full_plant_model <- lm(rel_fitness ~ growth_rate + days_to_flower +
    specific_leaf_area + root_shoot_ratio + chlorogenic_acid,
    data = plant_aridity)
```

(vi) Check the assumptions of your model. Do you need to transform any variables?

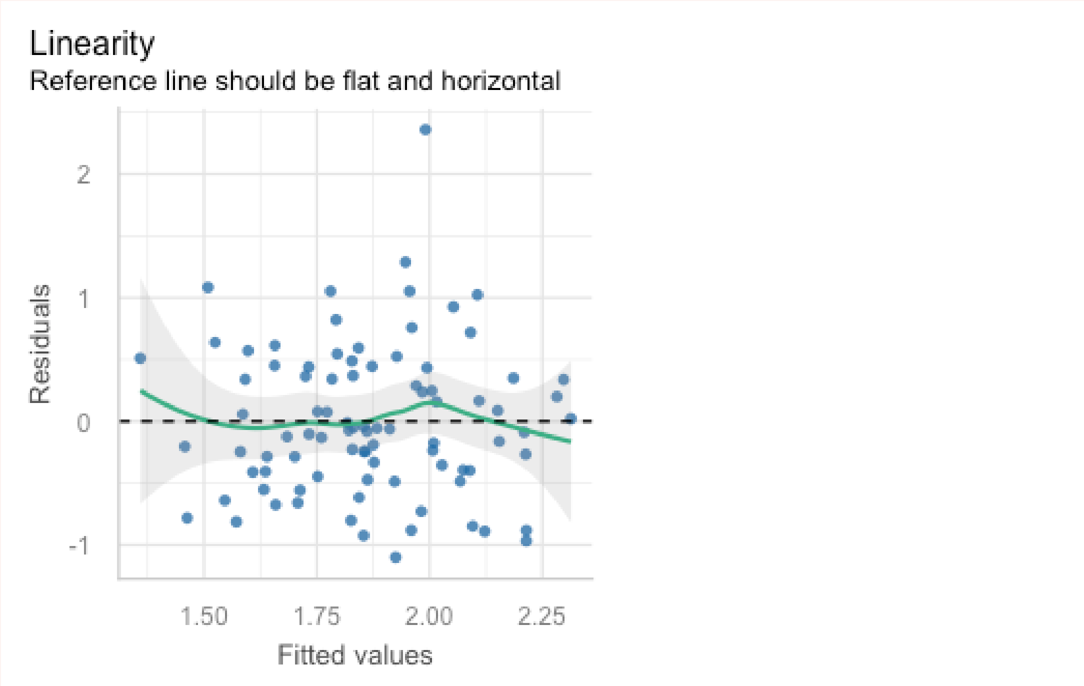
Hint 1: you can use the the `check_model()` function from the `performance` package to check the assumptions of your model.

Hint 2: If you are unsure about the collinearity plot, you can use the vif() function from the car package to get the VIF values for each predictor variable.

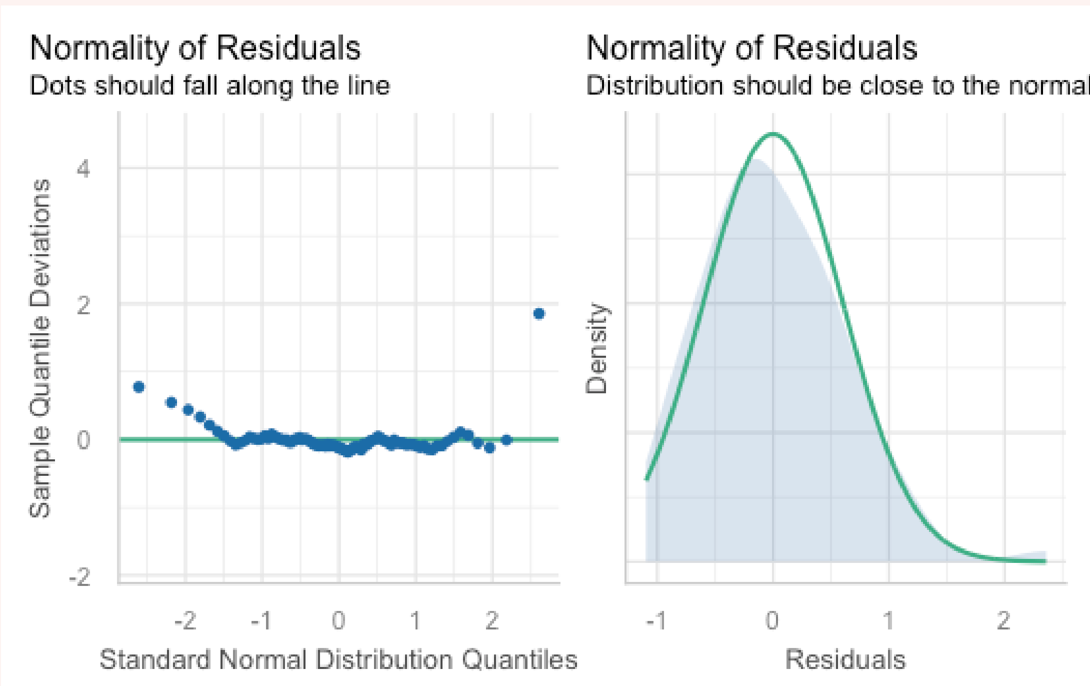
Hint 3: If you find that the assumptions are not met you can use log-transformations or other transformations as appropriate based on the distribution of your variables. If you are log-transforming a variable which contains negative or zero values you will need to add a constant, i.e. `log(x+1)`

CODE

```
check_model(full_plant_model)

# Individual plots:

# Linearity
check_model(full_plant_model, check = c("linearity"))
```

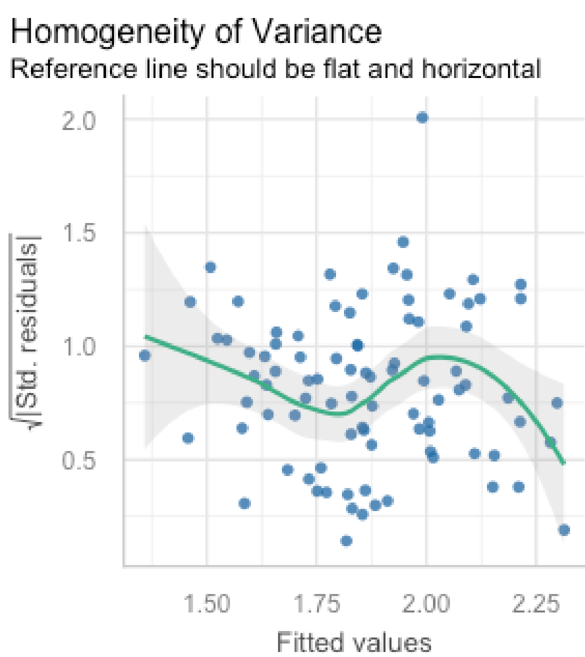



```
CODE
# Normality
check_model(full_plant_model, check = c("qq", "normality"))
```



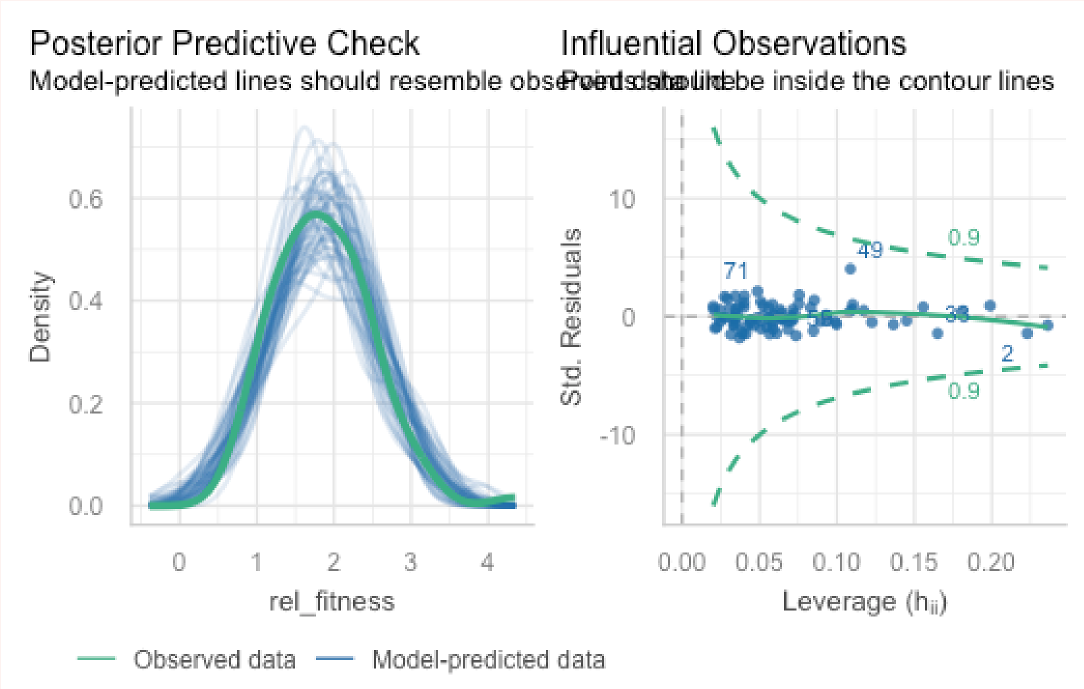
CODE

```
# Equal variance
check_model(full_plant_model, check = c("homogeneity"))
```



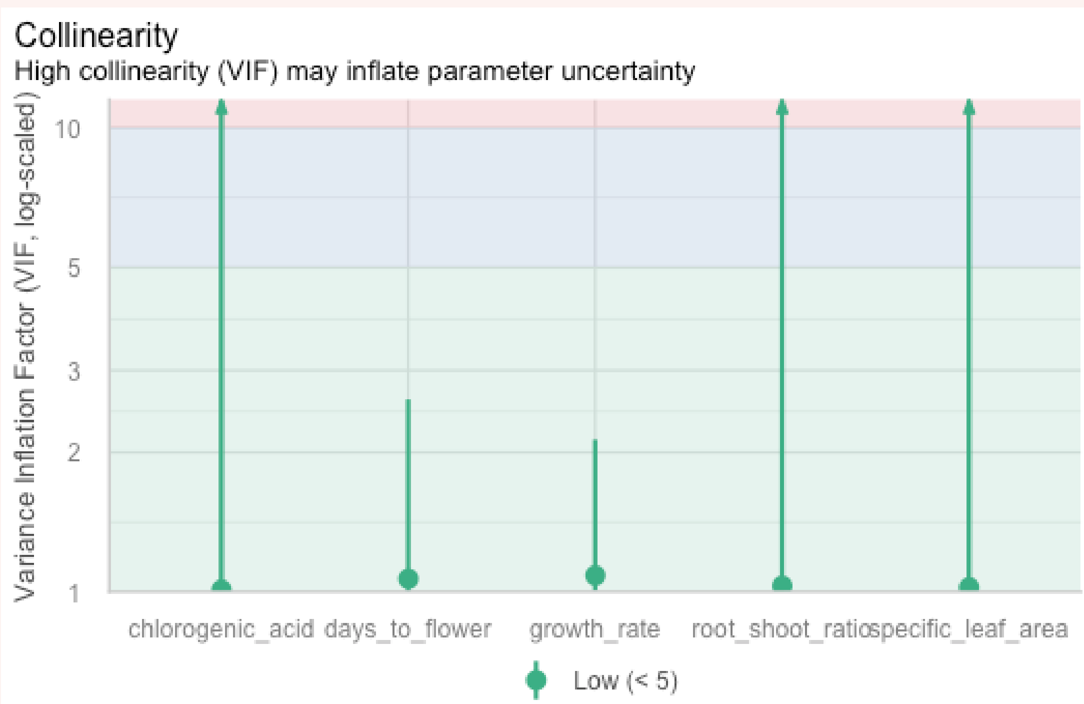
CODE

```
# Outliers
check_model(full_plant_model, check = c("outliers", "pp_check"))
```



CODE

```
# VIF - visual
plot(performance::check_collinearity(full_plant_model))
```



CODE

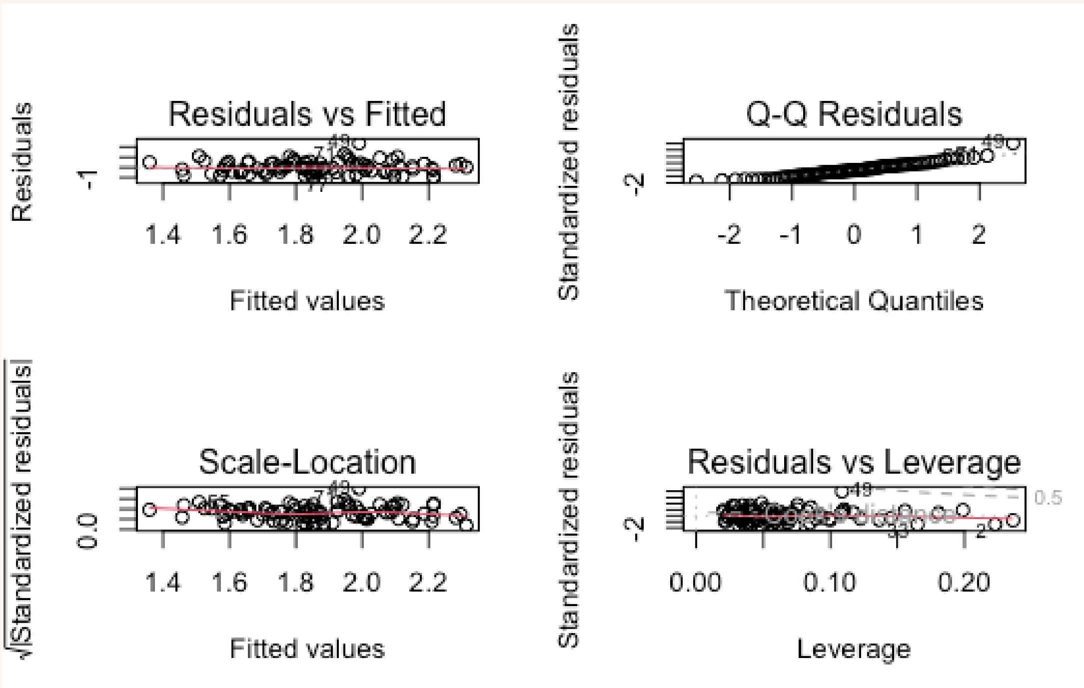
```r
# VIF - numbers
car::vif(full_plant_model)
```

OUTPUT

```
     growth_rate     days_to_flower specific_leaf_area    root_shoot_ratio
        1.085235           1.067528           1.022162            1.030175
 chlorogenic_acid
        1.010600
```

CODE

```r
### Alternately, you can use the base R plot function
par(mfrow=c(2,2))

plot(full_plant_model)
```



CODE

```r
par(mfrow=c(1,1))
```

Linearity: The residuals vs fitted plot shows a random scatter of points, suggesting that the linearity assumption is met.

Independence: The residuals appear to be randomly distributed (if correlated, they may follow a trend), which suggests that the independence assumption is met. We also know from the experimental design the observations are independant.

Normality: The Q-Q plot shows that the residuals are not normally distributed, as the points deviate from the straight line, especially in the tails producing a u-shape. This suggests that the normality assumption may be violated.

Equal variance: The scale-location plot shows a random scatter of points but the line is not too straight suggesting there may be some weird things happening.

Influential observations: The residuals vs leverage plot shows that there are no influential observations, as all points are within the Cook's distance lines.

Collinearity: No concern about collinearity as all VIFs are below 5.

Decision: We saw there were some issues with normality and equal variance, so we could try transforming the response variable to see if this improves the model assumptions.

CODE

```
full_plant_log_model <- lm(log(rel_fitness) ~ growth_rate + days_to_flower +
    specific_leaf_area + log(root_shoot_ratio+ 1) + chlorogenic_acid,
    data = plant_aridity)

check_model(full_plant_log_model)

### Alternately, you can use the base R plot function
par(mfrow=c(2,2))

plot(full_plant_log_model)

par(mfrow=c(1,1))
```

After further testing, we found that log-transforming the response variable improved the normality and equal variance of the residuals, but it was improved further by transforming root_shoot_ratio, which was highly skewed, so this was also transformed.

Herein we will use the transformed model.

(vii) Interpret the model parameters and fit statistics. Are there any significant predictors of relative fitness? does the model explain much variation in relative fitness?

CODE

```
summary(full_plant_log_model)
```

OUTPUT

```

Call:
lm(formula = log(rel_fitness) ~ growth_rate + days_to_flower +
    specific_leaf_area + log(root_shoot_ratio + 1) + chlorogenic_acid,
    data = plant_aridity)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
-0.7947 -0.2410  0.0169  0.2435  0.9532

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                0.49232    0.06291   7.826 1.44e-11 ***
growth_rate                0.10094    0.05400   1.869   0.0651 .
days_to_flower             0.01154    0.03645   0.317   0.7523
specific_leaf_area         0.06877    0.05757   1.194   0.2357
log(root_shoot_ratio + 1) -0.01304    0.03820  -0.341   0.7336
chlorogenic_acid          -0.08594    0.03631  -2.367   0.0203 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3431 on 83 degrees of freedom
Multiple R-squared:  0.1088,    Adjusted R-squared:  0.0551
F-statistic: 2.026 on 5 and 83 DF,  p-value: 0.08329
```

The overall model is not significant ($p > 0.05$), but we do see that chlorogenic_acid is significant and growth_rate is borderline significant. The adjusted R^2 value is 0.055, which means that approximately 6% of the variation in relative fitness can be explained by the model. This is a very low value, suggesting that the model does not explain much of the variation in relative fitness.

Although the model is not significant, let's see if there is a way to make it more parsimonious.

(viii) Remove any highly non-significant predictors ($P > 0.1$) from the model and run the model again. This will be known as the 'reduced model'.

CODE

```
reduced_plant_log_model <- lm(log(rel_fitness) ~ growth_rate + chlorogenic_acid,
    data = plant_aridity)


summary(reduced_plant_log_model)
```

OUTPUT

```

Call:
lm(formula = log(rel_fitness) ~ growth_rate + chlorogenic_acid,
    data = plant_aridity)

Residuals:
     Min       1Q   Median       3Q      Max
-0.78397 -0.25125  0.02644  0.23483  0.95917

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       0.51904    0.04517  11.490   <2e-16 ***
growth_rate       0.08618    0.05090   1.693   0.0941 .
chlorogenic_acid -0.08601    0.03587  -2.398   0.0186 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3406 on 86 degrees of freedom
```

```
Multiple R-squared:  0.0901,    Adjusted R-squared:  0.06894
F-statistic: 4.258 on 2 and 86 DF,  p-value: 0.01725
```

(ix) Run a partial f-test to determine whether the reduced model an improvement on the full model. Which model is better?

CODE

```
anova(full_plant_log_model, reduced_plant_log_model)
```

OUTPUT

```
Analysis of Variance Table

Model 1: log(rel_fitness) ~ growth_rate + days_to_flower + specific_leaf_area +
    log(root_shoot_ratio + 1) + chlorogenic_acid
Model 2: log(rel_fitness) ~ growth_rate + chlorogenic_acid
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     83 9.7718
2     86 9.9767 -3  -0.20485 0.58 0.6298
```

The p-value from the F-test is 0.6298, which is greater than 0.05, so we fail to reject the null hypothesis.

This means that the reduced model is not significantly different to the full model, and we would opt for the reduced model as it is more parsimonious.

(x) Run stepwise regression using the `step()` function fitted to the full model. Which model does it select as the best model?

CODE

```
step_model <- step(full_plant_log_model, direction = "backward")
```

OUTPUT

```
Start:  AIC=-184.61
log(rel_fitness) ~ growth_rate + days_to_flower + specific_leaf_area +
    log(root_shoot_ratio + 1) + chlorogenic_acid

                            Df Sum of Sq     RSS     AIC
- days_to_flower             1   0.01181  9.7836 -186.50
- log(root_shoot_ratio + 1)  1   0.01373  9.7855 -186.49
- specific_leaf_area         1   0.16798  9.9398 -185.10
<none>                                    9.7718 -184.61
- growth_rate                1   0.41143 10.1832 -182.94
- chlorogenic_acid           1   0.65952 10.4313 -180.80

Step:  AIC=-186.51
log(rel_fitness) ~ growth_rate + specific_leaf_area + log(root_shoot_ratio +
    1) + chlorogenic_acid

                            Df Sum of Sq     RSS     AIC
- log(root_shoot_ratio + 1)  1   0.00977  9.7934 -188.42
- specific_leaf_area         1   0.17558  9.9592 -186.92
<none>                                    9.7836 -186.50
- growth_rate                1   0.40165 10.1853 -184.93
- chlorogenic_acid           1   0.68071 10.4643 -182.52
```

```
Step:  AIC=-188.42
log(rel_fitness) ~ growth_rate + specific_leaf_area + chlorogenic_acid

                     Df Sum of Sq     RSS     AIC
- specific_leaf_area  1   0.18327  9.9767 -188.77
<none>                             9.7934 -188.42
- growth_rate         1   0.39195 10.1854 -186.92
- chlorogenic_acid    1   0.68699 10.4804 -184.38

Step:  AIC=-188.77
log(rel_fitness) ~ growth_rate + chlorogenic_acid

                   Df Sum of Sq     RSS     AIC
<none>                           9.9767 -188.77
- growth_rate       1   0.33248 10.3092 -187.85
- chlorogenic_acid  1   0.66707 10.6437 -185.01
```

CODE

```
AIC(full_plant_log_model, reduced_plant_log_model)
```

OUTPUT

```
                        df      AIC
full_plant_log_model     7 69.95815
reduced_plant_log_model  4 65.80463
```

The step() function has told us that the most appropriate model is the reduced model, as indicated by the smaller (more negative) AIC value. We can also see a similar outcome when we run the `AIC()` function for our models (70.0 for full model, 65.8 for reduced). This is consistent with our previous analysis using the F-test.

CODE

```
summary(step_model)
```

OUTPUT

```

Call:
lm(formula = log(rel_fitness) ~ growth_rate + chlorogenic_acid,
    data = plant_aridity)

Residuals:
     Min       1Q   Median       3Q      Max
-0.78397 -0.25125  0.02644  0.23483  0.95917

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       0.51904    0.04517  11.490   <2e-16 ***
growth_rate       0.08618    0.05090   1.693   0.0941 .
chlorogenic_acid -0.08601    0.03587  -2.398   0.0186 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3406 on 86 degrees of freedom
Multiple R-squared:  0.0901,	Adjusted R-squared:  0.06894
F-statistic: 4.258 on 2 and 86 DF,  p-value: 0.01725
```

In addition to the model becoming more parsimonious, we can also see that the model itself is now significant ($P<0.05$), however it still only explains 7% of variation in log(rel_fitness).

(viii) Write a concluding statement on your findings.

As identified by the partial F-test and stepwise regression, we found that the reduced model is most appropriate, which includes `growth  rate` and `chlorogenic_acid` as the predictors. This reduction in predictors has also made our model more significant, though it only explains 7% of variation in log(rel_fitness).

As our model does not explain much variation in the response, it is worth investigating whether another model would be better fit. In the original article they actually incorporated a treatment interaction into their model, so if you want to see how this was done, feel free to check out the article (source listed at the beginning of the exercise).

Conclusion

Closing thoughts

You have done a great job working to improve and produce more parsimonious models.

As you may have noticed in this lab, models are rarely perfect, which is okay; it is our job to find the best way to explain variation in our response variables by exploring alternatives.

Keep up the great work and see you in the week 9 lecture for the last week of modelling!

If you have any questions, please approach your tutors or you are welcome to post to the Ed Discussion board.

Attribution